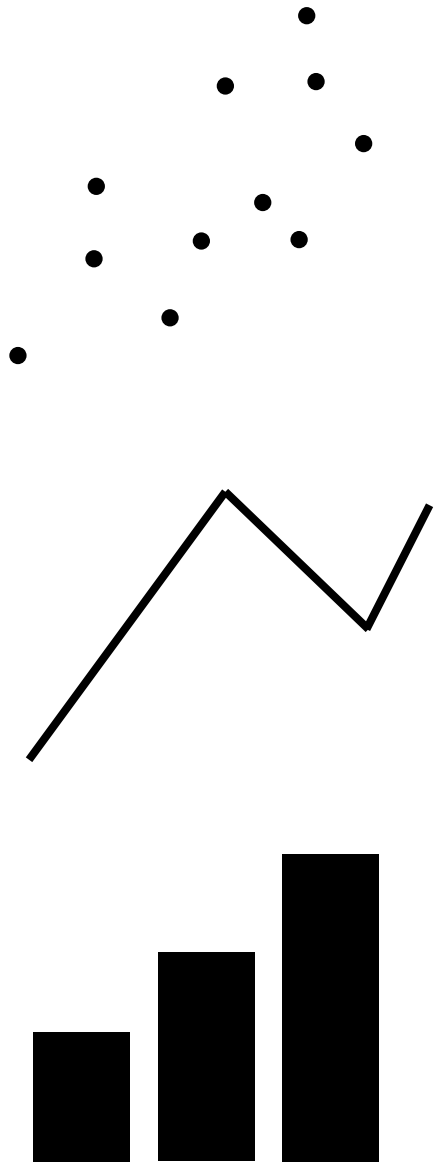
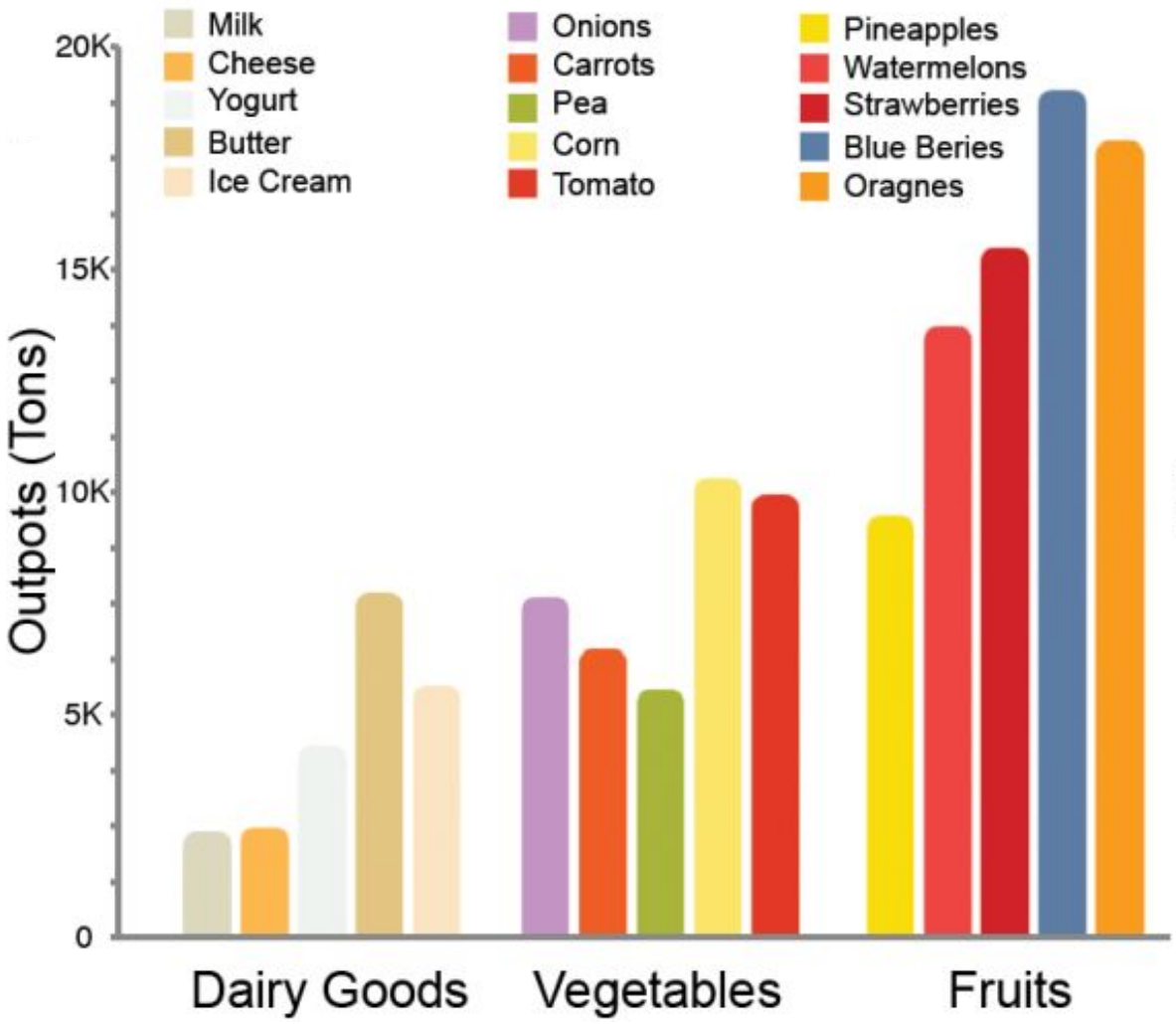
Do You See What I See?

A Qualitative Study Eliciting High-Level Visualization Comprehension

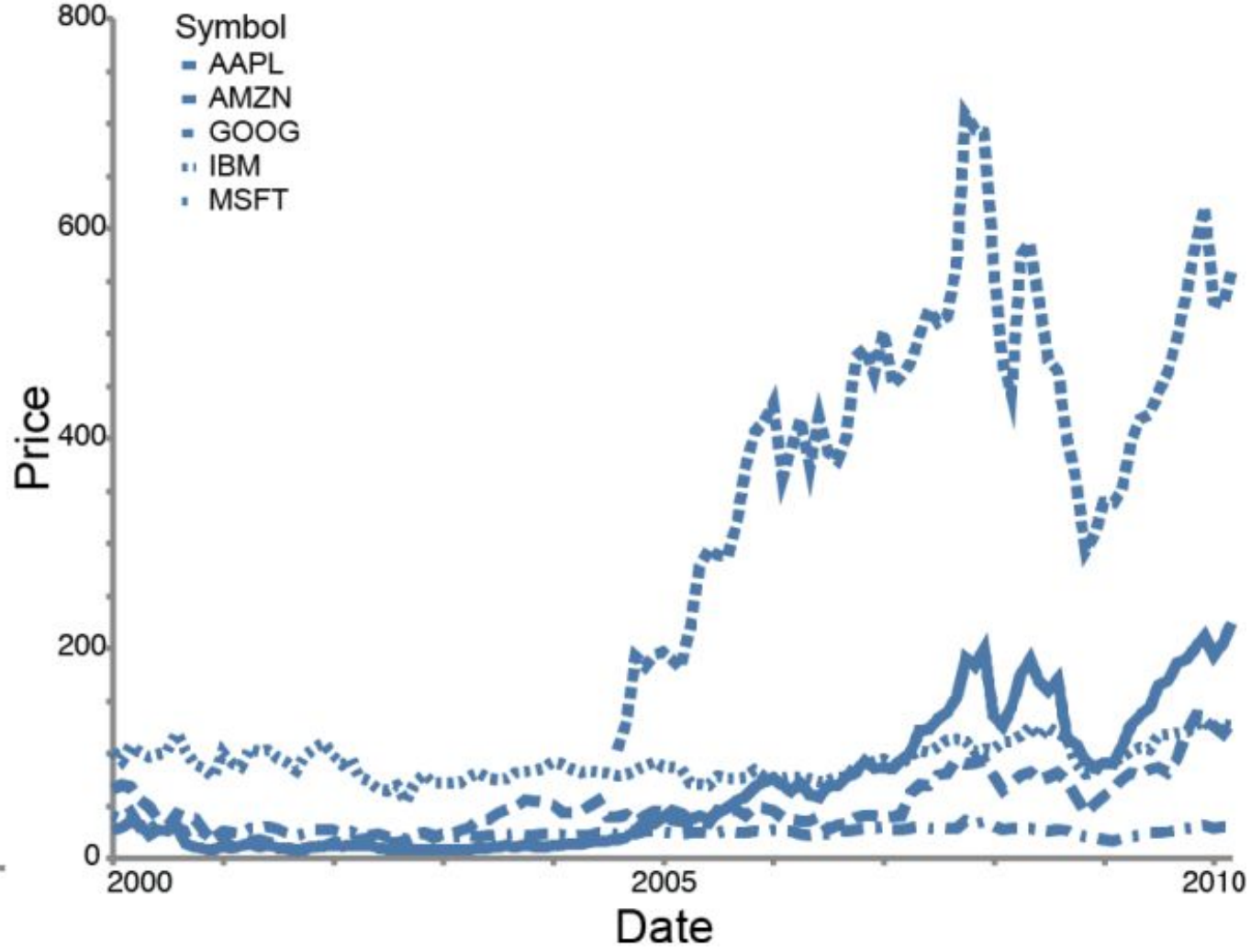


What is your interpretation of these two graphs?

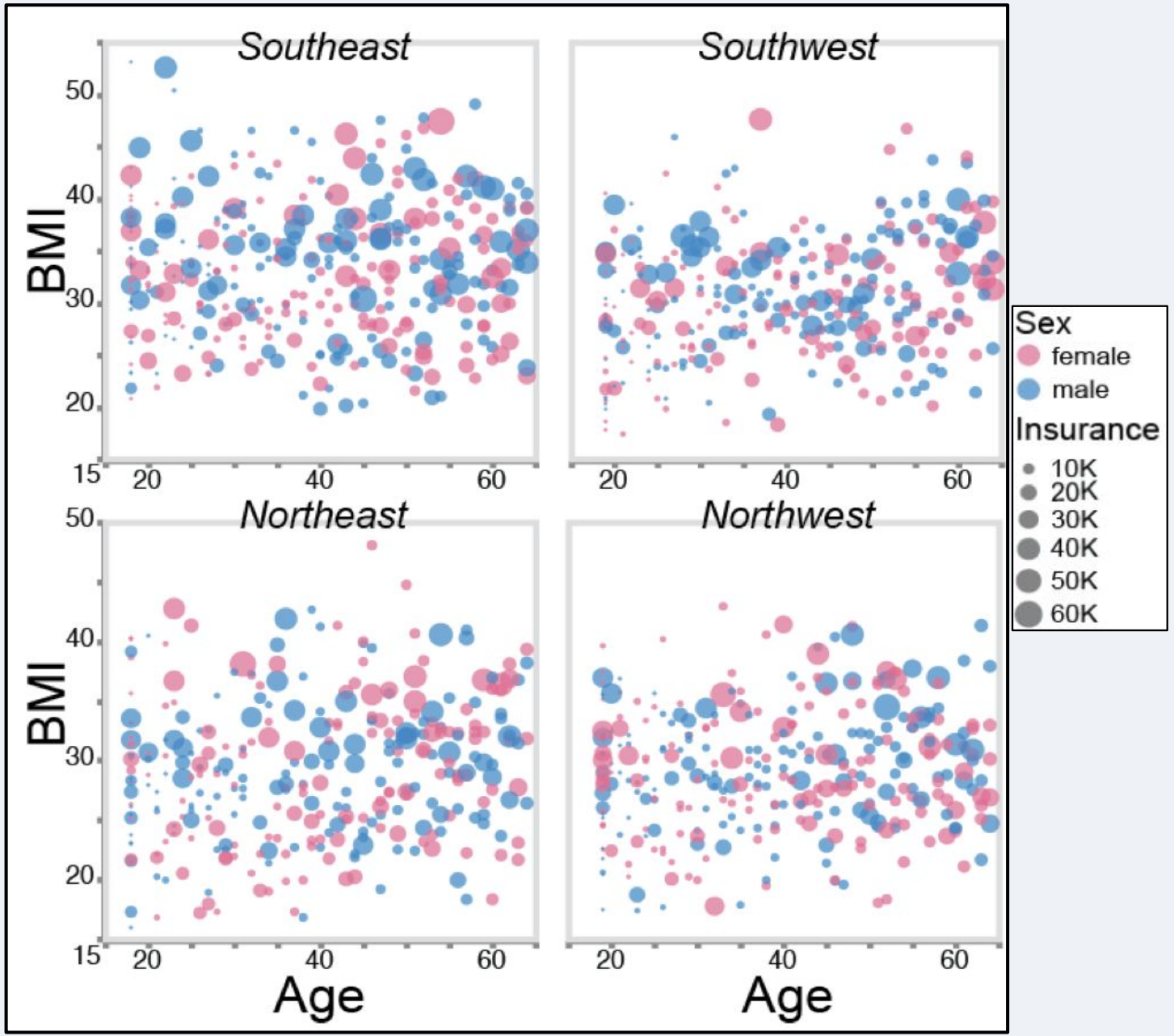
3/5 ☺ 0/5 😐 2/5 ☹



2/5 ☺ 1/5 😐 2/5 ☹



The problem: Designers have a message. Do viewers get it?



Two ways people attend to a chart

Top-down (goal-driven)

The reader is told exactly what to find:

Estimate the correlation between IMDB rating and gross movie sales.

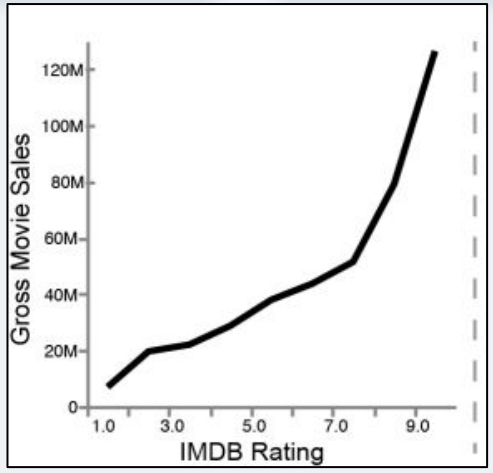
classic task

Bottom-up (stimulus-driven)

The reader gets one instruction, and no hint of a target statistic:

Describe what you see in the graph.

this paper's task



What the study intend to answer

RQ1: Alignment

Do the patterns people see in a visualization match the visualization's stated objective?

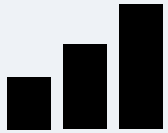
RQ2: Discovery

What types of patterns and statistics do people naturally see in common visualization designs, unprompted?

Tested across these 3 chart types



Scatterplot



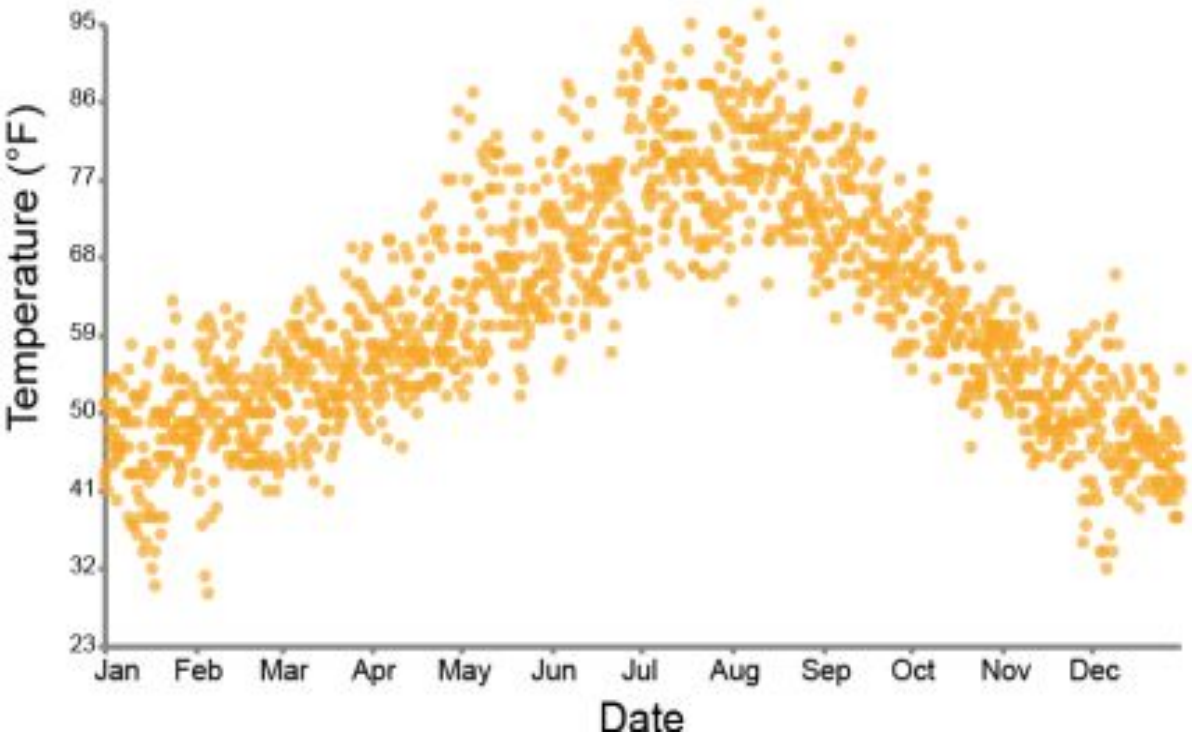
Bar graph



Line graph

One chart, two different reads

“Sunny day” chart, max. temperature by date



Designer's intent: *show the days sunny weather occurs, and how the max. temperature rises through summer and falls again.*

General Match (GM)

Changes in temperature across the year and particular weather characteristics of each day. The temperature starts low in January, goes up until August, and goes down from there.

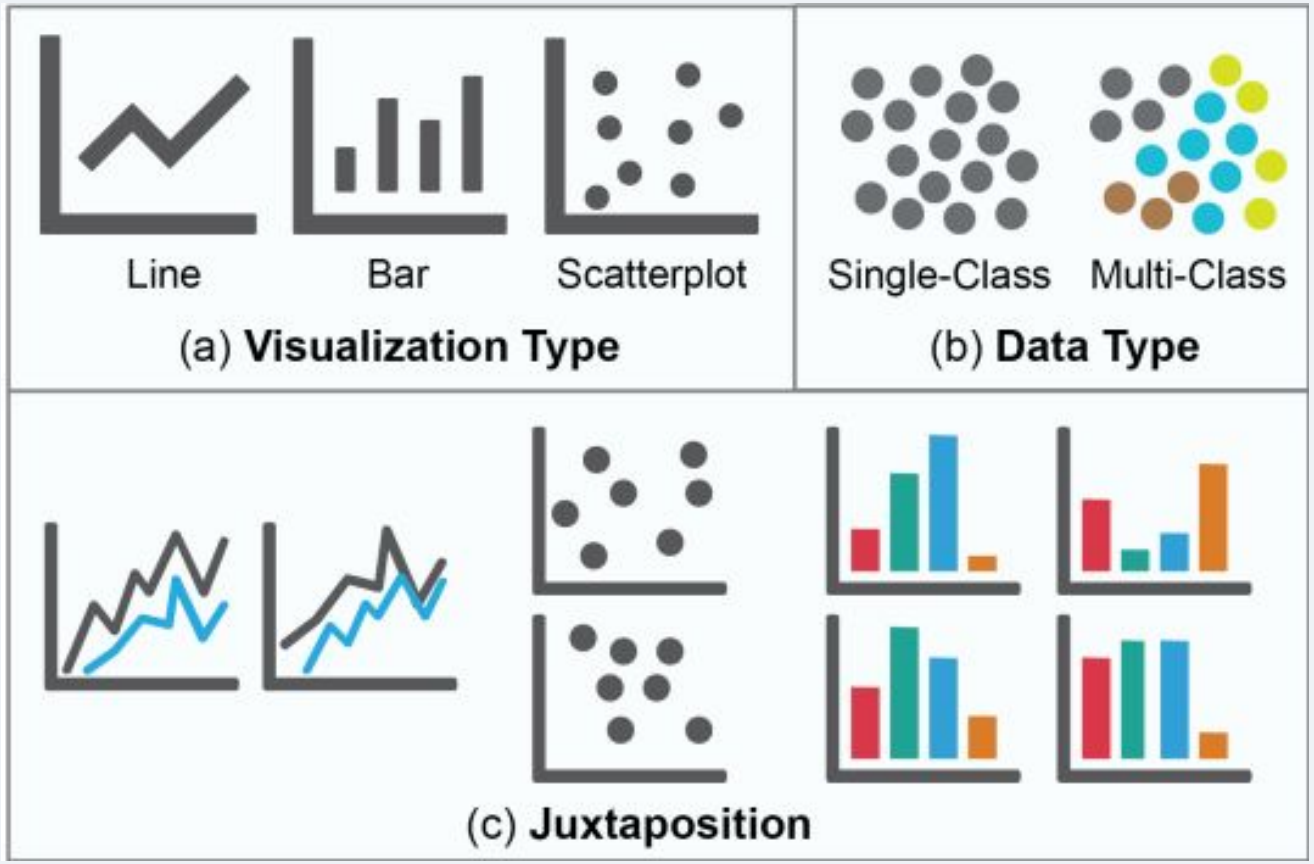
Partial Match (PM)

The graphs describe the temperature of different months in 2012 under the different weather.

Matches aligned at one of four thresholds:
Complete Match (CM), General Match (GM), Partial Match (PM), No Match (NM)

Method

3 chart types × 2 data types × 2 compositions = 12 different conditions



Each of the paper's charts is a real design , replotted onto a new dataset and stripped of titles and captions, so only the visualization itself could drive comprehension.

24 participants, each describing a run of charts

Every participant saw two prompts per chart, both deliberately free of any statistical leads:

TASK 1

Describe what you see in the graph.

broad, contextualized insight

TASK 2

What questions would you be able to answer from this graph?

nudges toward analysis, without naming a statistic

24 participants each worked through 12 charts, one from every condition in the study design.

Response Coding	Scatterplot					Line Graph					Bar Graph					All
	SC	MC	SC-J	MC-J	All	SC	MC	SC-J	MC-J	All	SC	MC	SC-J	MC-J	All	
Complete Match	12	11	8	8	39	12	8	11	9	40	11	8	12	7	38	117
General Match	3	3	1	2	9	3	0	1	1	5	3	5	1	3	12	26
Partial Match	4	6	8	5	23	4	7	3	8	22	4	7	6	5	22	67
No Match	5	4	7	9	25	5	9	9	6	29	6	4	5	9	24	78
All	24	24	24	24	96	24	24	24	24	96	24	24	24	24	96	288

THEME 1

Intention does not align with comprehension

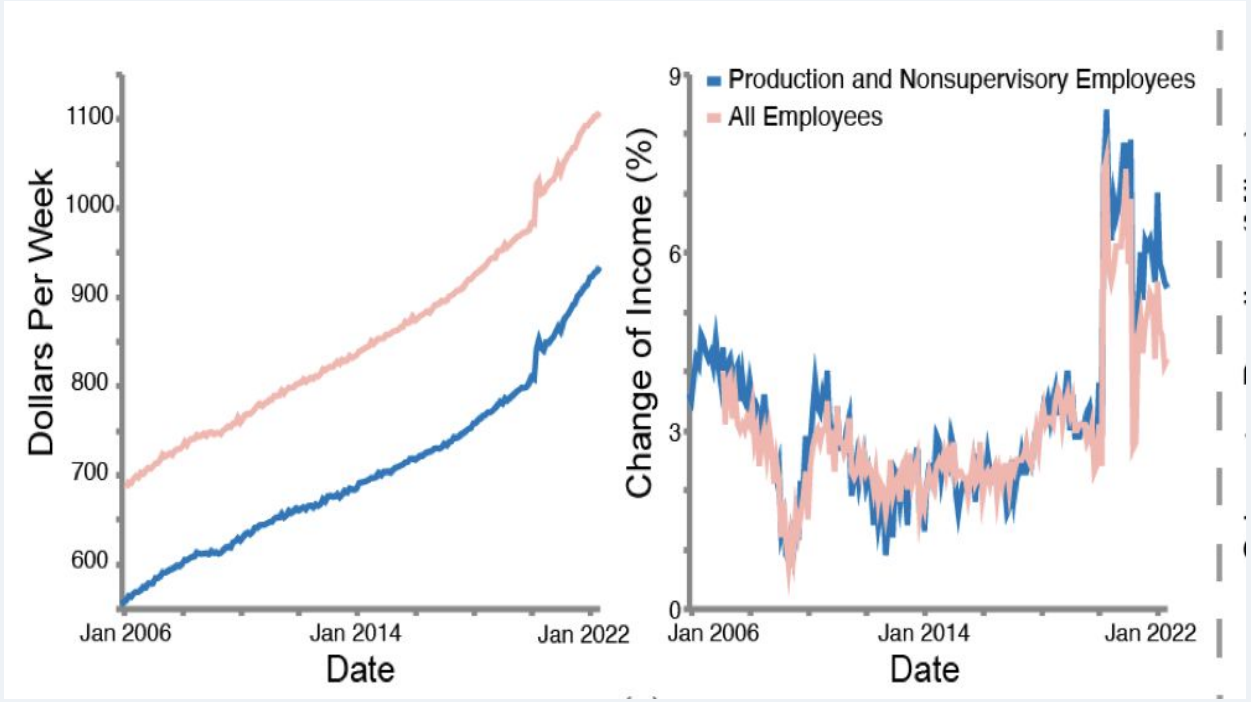
It is easier to read bar charts and simple line graphs but not scatterplots or graphs involving multiple categories.

There are two graphs. From the right one, I see their income change over the years. I am seeing two line graphs showing the average weekly earnings of 2 categories of employees… and their change of income over the previous year.

In the first chart four BMI-vs-age panels by region, only one of three viewers who described it named the actual message (no clear correlation). The other two just re-stated the axis labels.

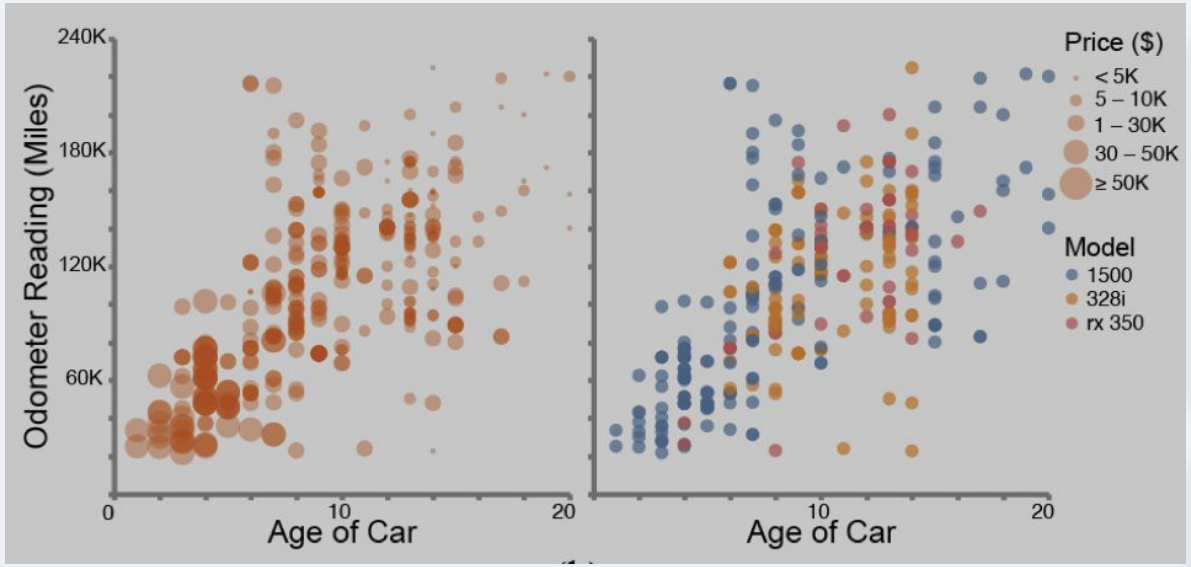


The chart was designed to show a comparison between two employee groups' pay growth. This person described each panel separately, and never made the comparison the chart was built for.



THEME 2

Cued-task ability ≠ what people notice unprompted



Scatterplots are well known to make correlation, clusters, and distribution easy to spot which was what this chart was built to show.

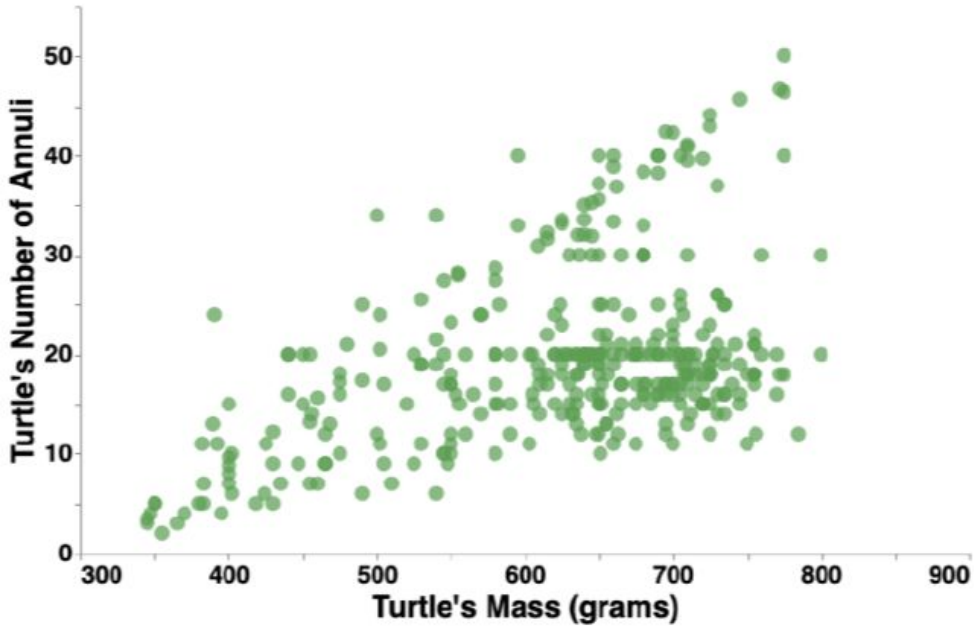
I am seeing a dot graph showing different models of cars with different price points measured by their odometer reading at a specific age. I can tell how many miles a specific car has based on age.

Key telling the price of the car and the model of the car. The age of the car and the odometer reading in miles.

Chart type alone doesn't predict what people extract

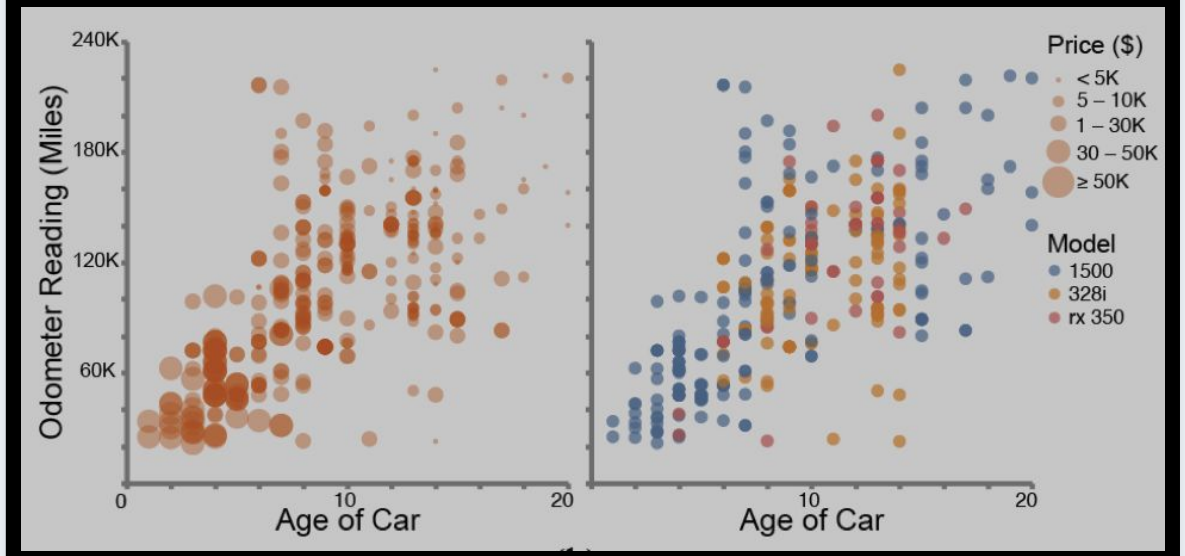
Same chart type, same underlying question, composition changes thing

One chart, one message



Easier to read, shows the correlation between the turtle's body mass and flipper length per species in individual graphs.

Split into side-by-side panels



In the left, I see plots showing the cost of a car and how old the car is… On the right, I see three different colored graphs. I am assuming they are showing different car models.

WHAT READERS STRUGGLED WITH

Design & data critiques that came up unprompted

Unfamiliar jargon

“I don't know what IMDB is.”

“I do not follow stocks or computer programming types.”

Confusing encodings

“Difficult to distinguish stroke dash, making the graph confusing.”

Too much complexity

“The messier and more colorful, the more my brain shuts down and doesn't want to process it.”

When it worked: semantic color

“Output of specific foods, grouped by their food group. Almost all of the fruits had higher production than anything else, while dairy had the lowest.”

Four complaints that showed up again and again regardless of chart type, plus one example of a design choice that worked.

IMPLICATIONS

How you ask changes what you find

For designers

- Simplicity beats expressiveness, packing more into one chart often communicates less reliably
- Scaffolding helps: annotations, baselines and clear legends measurably raised alignment
- Chart-type ⟶ task mapping tools also need to account for composition and data complexity

Same participant pool, two different prompts, two different-looking results.

For evaluation methods

“What question could this graph answer?

How do the sales of item X change throughout the year and the months?

an example participant question

vs. the same chart, “Describe this graph”:

This graph has age of car and odometer reading on both graphs there difference price and age and model information

LIMITATIONS & FUTURE WORK

Where this preliminary study leaves off

Narrow chart set

Only scatterplots, bar and line graphs. No maps, pie charts, or bubble charts yet.

Unsystematic sampling

Didn't control for education, profession, or visualization literacy across participants.

Self-report only

No eye-tracking or recall data, can't see WHERE attention actually goes.

Narrative stimuli only

All charts had a single “correct” intended message. Not exploratory, open-ended analysis.

TAKEAWAYS

Seeing isn't the same as being shown

1

Intent ≠ comprehension

Even the BMI chart, built with best practices, mostly got read as a list of axis labels.

2

Cued ≠ natural

The car-models scatterplot invites correlation but viewers described dots and labels, not the relationship.

3

Chart type isn't enough

The turtle chart read cleanly as one plot; splitting the same idea into panels made it shallow.